BioSpecInfo

Software Bill of Materials (SBOM)

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Versione **bsi-v237** · commit `7dc3f01`

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Documento generato automaticamente dal sorgente Markdown versionato nel repository. La versione e il commit stampati in ogni piè di pagina identificano esattamente lo stato del software descritto.

Software Bill of Materials (SBOM)

BioSpecInfo · document generated by `tools/genera-sbom.js`.

Application version	`bsi-v237`
Commit	`7dc3f011df8f8c0646f9eda186923f345f1ca23b`
Generated (UTC)	`2026-10-29T08:48:29.952Z`
Machine-readable format	`../evidence/sbom.cdx.json` — CycloneDX 1.5

> The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

Component	Version/File	Licence	Size	Role
RDKit MinimalLib (JS glue) _{RDKit / Greg Landrum}	`RDKit_minimal.js`	BSD-3-Clause	125 kB	Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing.
RDKit MinimalLib (WebAssembly) _{RDKit / Greg Landrum}	`RDKit_minimal.wasm`	BSD-3-Clause	6753 kB	WebAssembly binary of the cheminformatics engine.
three.js _{mrdoob e collaboratori}	`three.min.js`	MIT	589 kB	WebGL rendering engine for the 3D scenes.
three.js — UnrealBloomPass _{mrdoob e collaboratori}	`three_bloom.js`	MIT	25 kB	Post-processing effect for the astronomical visualisations.
three.js — GLTFLoader _{mrdoob e collaboratori}	`gltf_loader.js`	MIT	94 kB	Loading of anatomical and molecular models in glTF format.
3Dmol.js _{University of Pittsburgh / David Koes}	`3Dmol-min.js`	BSD-3-Clause	502 kB	3D molecular visualisation (PDB, SDF, surfaces).

Component	Version/File	Licence	Size	Role
SmilesDrawer _{Daniel Probst}	`smiles-drawer.min.js`	MIT	246 kB	2D structure drawing from SMILES.
PDF.js (build legacy) _{Mozilla Foundation}	`vendor/pdfjs/pdf.min.mjs`	Apache-2.0	511 kB	Opening PDF documents in the spectrum reader: every page drawn and its text layer. It is the legacy build, because the default one uses very recent language features and breaks on browsers that do not have them yet.
PDF.js — worker _{Mozilla Foundation}	`vendor/pdfjs/pdf.worker.min.mjs`	Apache-2.0	1290 kB	The PDF decoding work, off the main thread: a complex page does not freeze the interface.
sql.js (JS glue) _{sql.js / SQLite}	`lib/sql-wasm.js`	MIT	45 kB	In-browser SQLite database for the data exercises.
SQLite (WebAssembly) _{SQLite Consortium}	`lib/sql-wasm.wasm`	Public Domain	644 kB	SQLite engine compiled to WebAssembly.
Dataset dimuoni (CERN Open Data) _{CERN Open Data Portal}	`lib/dimuon.js`	CC0-1.0	48 kB	Experimental data for the particle-physics exercise.
Modello 3D «Noir» (Skyfox13, Sketchfab) _{Skyfox13 — https://sketchfab.com/podderkuntal25}	`models/spectra-noir.glb`	CC-BY-4.0	8154 kB	Spectra's face in the Spectra Live console. Modified: textures recompressed, tangents removed, face in dark grey metal, mouth and head animated. Credited in the console.

Digests

File	SHA-256
RDKit_minimal.js	58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9
RDKit_minimal.wasm	e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8
three.min.js	9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2
three_bloom.js	42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090
gltf_loader.js	5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6
3Dmol-min.js	06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff
smiles-drawer.min.js	955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2
vendor/pdfjs/pdf.min.mjs	bccc24ea711db8e44503629519904a5292d73b9daaa214bbe7cdcc282b0f4259
vendor/pdfjs/pdf.worker.min.mjs	145d2dd3ab0c86151011dba95acfa2d5336e2accd59388ea43dbee0efddaaec6
lib/sql-wasm.js	77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f
lib/sql-wasm.wasm	438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed
lib/dimuon.js	61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d
models/spectra-noir.glb	b781b54819d1d2aabf751788a813d8da830dabaddb913085352b572140903479

2. Code written for this project

File	Size	Role
index.html	4700 kB	Main application: interface, teaching sections, chemical data.
bsi-ai-hub.js	432 kB	«Spectra» AI agent: agentic loop, tools, provider management.
bsi-spettri.js	35 kB	IR/NMR spectral prediction engine over the molecular graph.
bsi-cheminfo.js	125 kB	Cheminformatics engine: standardisation, fingerprints, clustering, PCA, QSAR models with a null model.
bsi-pretsch.js	37 kB	The NMR estimation tables transcribed in full from Pretsch–Bühlmann–Badertscher, 4th ed.: ^{13}C and ^{1}H benzenes, ethylenes, alkanes, alkynes, ^{13}C aliphatics, steric corrections, ^{1}J(C,H). Numbers only, no code.
bsi-nmr.js	57 kB	Per-atom ^{1}H and ^{13}C NMR prediction: applies the bsi-pretsch.js tables by walking the ring and measuring distances, with a cyclic reference compound and chemical equivalence by environment code.
bsi-geom3d.js	46 kB	3D coordinates from a SMILES by distance geometry, carrying the NMR predictor's own atom indices: the precondition for the peak↔atom↔3D link.
bsi-fisica-comune.js	31 kB	The shared home of the physics sections: header, tabs, sliders, canvas charts, language.
bsi-fis-schrodinger.js	22 kB	1D Schrödinger equation: finite differences, Sturm bisection, inverse iteration, Crank-Nicolson, Morse for real molecules.
bsi-fis-hartreefock.js	29 kB	Restricted Hartree-Fock in the STO-3G basis: McMurchie-Davidson integrals, Boys function, SCF with DIIS, molecular orbitals.

File	Size	Role
`bsi-fis-rotovib.js`	16 kB	IR and Raman rovibrational spectra of diatomics from molecular constants; combination differences.
`bsi-fis-termostat.js`	11 kB	Statistical thermodynamics: partition functions, entropy, heat capacity, Boltzmann populations.
`bsi-fis-dinamica.js`	12 kB	2D Lennard-Jones molecular dynamics: velocity Verlet, thermostat, virial, radial distribution.
`bsi-fis-matematica.js`	24 kB	Mathematics for chemistry: FFT of the FID, Hückel, Runge-Kutta for kinetics, least squares with errors.
`bsi-fis-cristalli.js`	20 kB	Crystallography: Bravais lattices, Miller indices, powder pattern with structure factor and absences.
`bsi-fis-nucleare.js`	14 kB	Radiochemistry: Bateman equations, radiometric dating, Weizsäcker semi-empirical formula.
`bsi-mat-cas.js`	67 kB	Computer-algebra engine: parser, exact fractions, canonical form, derivatives, verified antiderivatives, truncated series, limits.
`bsi-mat-basi.js`	26 kB	Mathematics basics: significant figures, logarithms and pH, equations with working, practice.
`bsi-mat-funzioni.js`	17 kB	Function analysis: domain, zeros, extrema, inflections, asymptotes; families and transformations.
`bsi-mat-calcolo.js`	21 kB	Derivatives with rules, integrals and Riemann sums, Taylor polynomials, limits.
`bsi-mat-lineare.js`	38 kB	Exact-fraction linear algebra: step-by-step Gauss-Jordan, determinant, inverse, eigenvalues, balancing as a null space.
`bsi-mat-ode.js`	21 kB	Differential equations: direction field, Euler/Heun/RK4, phase plane and equilibria from the Jacobian.
`bsi-mat-multi.js`	22 kB	Functions of several variables: surfaces, critical points from the Hessian, Maxwell relations, exact differentials.
`bsi-mat-complessi.js`	20 kB	Complex numbers (Argand plane, roots, Euler) and Fourier series.
`bsi-mat-risolutore.js`	110 kB	Mathematics exam solver: step-by-step working for 16 exercise types.
`bsi-mat-esame.js`	56 kB	Istituzioni di Matematiche I and II exam: syllabus, solver, solved papers, simulation.
`bsi-fis-esame.js`	135 kB	General Physics I and II exam: syllabus, 48 problem models, solved papers.
`bsi-percorsi.js`	9 kB	Study paths from high school to PhD for chemistry, physics and mathematics.
`bsi-mat-probabilita.js`	18 kB	Probability in motion: Galton, central limit, Monte Carlo, Bayes.
`bsi-mat-visiva.js`	24 kB	Visual mathematics: linear maps, Fourier epicycles, fractals, logistic map.
`bsi-fis-em.js`	21 kB	Electromagnetism in motion: charges, induction, generator, EM waves.
`bsi-fis-suono.js`	18 kB	Sound and acoustics with Web Audio: harmonics, timbre, Doppler, decibels.
`bsi-fis-solido.js`	16 kB	Semiconductors: Fermi-Dirac, doping, p-n junction, LEDs.
`bsi-chi-distillazione.js`	17 kB	Distillation: Antoine, van Laar, McCabe-Thiele, Rayleigh.
`bsi-chi-voltammetria.js`	12 kB	Simulated cyclic voltammetry: diffusion and Butler-Volmer.

File	Size	Role
`bsi-mat-numeri.js`	16 kB	Numbers and cryptography: sieve, Euclid, modular arithmetic, RSA, prime distribution.
`bsi-mat-grafi.js`	22 kB	Graphs and optimisation: Dijkstra, Kruskal, gradient descent, linear programming.
`bsi-mat-serie.js`	32 kB	Sequences and series: ε–N limit, tests, rearrangement, radius of convergence.
`bsi-mat-variazioni.js`	38 kB	Calculus of variations: brachistochrone, catenary, catenoid, Fermat.
`bsi-mat-topologia.js`	43 kB	Topology: Euler characteristic, surfaces, fundamental polygons, knots.
`bsi-mat-algebra.js`	45 kB	Algebraic structures: dihedral groups, permutations, finite fields, Lagrange.
`bsi-mat-giochi.js`	36 kB	Game theory: Nash, iterated dilemma, minimax, Nim.
`bsi-mat-logica.js`	46 kB	Logic and computability: truth tables, circuits, Turing machine, sets.
`bsi-mat-inferenza.js`	39 kB	Statistical inference: intervals, tests and power, regression, bootstrap.
`bsi-mat-noneuclidea.js`	48 kB	Non-Euclidean geometries: sphere, Poincaré disk, curvature and parallel transport.
`bsi-fis-gravita.js`	18 kB	Gravitation: N bodies with Verlet, Lagrange points, gravity assist.
`bsi-fis-calore.js`	17 kB	Heat and diffusion: 1D and 2D heat equation, Newton cooling, walls.
`bsi-fis-ising.js`	14 kB	Ising model: Metropolis, phase transition, hysteresis.
`bsi-fis-particelle.js`	40 kB	Particle physics: Standard Model, conservation, invariant mass, lifetimes.
`bsi-fis-analitica.js`	40 kB	Analytical mechanics: phase space, double pendulum, Noether, Liouville.
`bsi-fis-laser.js`	34 kB	Lasers and quantum optics: threshold, cavity modes, Rabi, Mach–Zehnder.
`bsi-fis-clima.js`	34 kB	Climate physics: radiative balance, CO2 forcing, ice-albedo, Coriolis.
`bsi-chi-fasi.js`	20 kB	Phase diagrams: water and CO2 (IAPWS, Span-Wagner), eutectic, Clapeyron.
`bsi-chi-pes.js`	17 kB	LEPS potential energy surface for H + H2 and trajectories.
`bsi-articoli-nobel-fisica.js`	118 kB	Topic explanations of the Nobel Prizes in Physics (no data).
`bsi-articoli-premi-matematica.js`	123 kB	Topic explanations of the Abel Prize and Fields Medal (no data).
`bsi-materie.js`	11 kB	Subject switch buttons: chemistry, physics, mathematics.
`bsi-fis-astro.js`	36 kB	Astrophysics: H-R diagram, black body, solar system, exoplanets, cosmology.
`bsi-fis-relativita.js`	15 kB	Special relativity: Minkowski, twins, dynamics.
`bsi-materie-app.js`	15 kB	Three apps in one: the chosen subject adapts the ✨ menu, Spectra, Study and Utility.
`bsi-attrezzi.js`	23 kB	Quick tools of the ✨ menu for physics and mathematics: calculator, derivative, graph, constant, units, formula, glossary, quiz.
`bsi-studio.js`	66 kB	Study-kit engine: summaries, quizzes, Leitner flashcards, formula sheet, glossary.
`bsi-studio-fisica.js`	306 kB	Physics study-kit data (no code).
`bsi-studio-matematica.js`	290 kB	Mathematics study-kit data (no code).
`bsi-risorse-materie.js`	149 kB	Web resources, bibliography and citations for the three subjects; CODATA constants and units; mathematical tables with their proofs.

File	Size	Role
`bsi-astro-sistemasolare.js`	110 kB	Solar system 3D at any date: Standish JPL elements, Kepler, Earth, Moon and eclipses, Jupiter and Saturn.
`bsi-astro-orbite.js`	69 kB	Orbital mechanics 3D: elements and state, manoeuvres, satellites, Lambert.
`bsi-astro-matcelesti.js`	60 kB	Celestial mathematics: Kepler equation, rotations and quaternions, N-body integrators, periodograms.
`bsi-astro-stelle.js`	63 kB	Life of stars 3D: Lane-Emden, evolution, nucleosynthesis.
`bsi-astro-buchineri.js`	72 kB	Black holes and general relativity: Schwarzschild ray tracing, geodesics, curvature, Hawking.
`bsi-astro-ondegrav.js`	55 kB	Gravitational waves: inspiral, chirp, detector, Hulse-Taylor pulsar.
`bsi-astro-galassie.js`	70 kB	Galaxies 3D: Hubble types, rotation curves, collision, Local Group.
`bsi-astro-cosmo.js`	70 kB	Cosmology 3D: Friedmann-Lemaître, distances, CMB, thermal history.
`bsi-astro-cielo.js`	75 kB	Planetarium 3D: the sky from any place and time, coordinates, precession, analemma.
`bsi-astro-esopianeti.js`	57 kB	Exoplanets 3D: habitable zone, transit, radial velocity, direct imaging.
`bsi-fis-spin.js`	55 kB	Spin, qubits and entanglement: Stern-Gerlach, Bloch sphere, CHSH, BB84.
`bsi-fis-statistica.js`	47 kB	Statistical mechanics: Schottky, Maxwell-Boltzmann, Fermi and Bose, Einstein and Debye.
`bsi-fis-nuclei.js`	49 kB	Nuclei and reactions: valley of stability, Q values, tunnelling, 3D nucleus, reactor.
`bsi-fis-materiali.js`	48 kB	Elasticity and materials: tension, beams, Euler, sound in solids.
`bsi-fis-elettronica.js`	48 kB	Electronics: rectifiers, diodes and Zener, common emitter, op-amps, ADC and DAC.
`bsi-mat-edp.js`	75 kB	Partial differential equations: string, membrane, heat, Laplace, Burgers.
`bsi-mat-analisicomplessa.js`	70 kB	Complex analysis: domain colouring, conformal maps, Joukowski, residues.
`bsi-mat-markov.js`	65 kB	Markov chains and stochastic processes: PageRank, gambler's ruin, walks, queues.
`bsi-mat-svd.js`	58 kB	SVD and applied linear algebra: 2×2 and 3×3 SVD, compression, PCA, least squares.
`bsi-mat-codici.js`	59 kB	Codes and elliptic curves: Hamming, CRC, curves over ℝ and F_p, ECDH.
`bsi-spectra-live.js`	71 kB	Spectra Live: voice console, local commands, screen reading, nine agent tools.
`bsi-spectra-androide.js`	33 kB	Spectra's face on the «Noir» model: animated head and mouth, 17 emotions from 11 expressive fields.
`bsi-spectra-mente.js`	17 kB	Spectra's mind: local memory, internal state, thoughts (a simulation of behaviour).
`bsi-spectra-robot.js`	20 kB	Fallback face: android head built in three.js.
`bsi-spectra-volto.js`	16 kB	Last-resort face: SVG vector portrait.
`bsi-aggiorna.js`	5 kB	«Check for updates»: version in use from the service worker, published version from sw.js.
`bsi-fis-fluidi.js`	15 kB	Fluids: Venturi and Bernoulli, viscosity and Stokes, capillarity.
`bsi-mat-vettoriale.js`	23 kB	Vector calculus: fields, Green, gradient.
`bsi-chi-cinetica.js`	17 kB	Complex kinetics: composite and oscillating reactions, Turing patterns.

File	Size	Role
bsi-fis-meccanica.js	23 kB	Mechanics in motion: projectile with air, collisions, real pendulum, coupled oscillators.
bsi-fis-onde.js	20 kB	Waves and optics: superposition, lenses and mirrors, diffraction.
bsi-fis-circuiti.js	17 kB	Circuits: RLC transients, alternating current, filters and Bode.
bsi-fis-termo.js	25 kB	Thermodynamics in motion: cycles, particle gas, Brownian motion.
bsi-mat-numerici.js	24 kB	Numerical methods: roots, integration, ODEs, interpolation with measured orders.
bsi-mat-geometria.js	24 kB	Analytic geometry and vectors: vectors, lines and planes in 3D, conics.
bsi-chi-equilibrio.js	13 kB	Chemical equilibrium and Le Chatelier: kinetic simulator and K(T).
bsi-lab3d.js	63 kB	3D Laboratory: surfaces,
bsi-vista-molecole.js	8 kB	Quality settings and control bar for every 3Dmol molecular viewer.
bsi-assistente.js	71 kB	The three macro-areas Chemistry, Physics and Mathematics: natural-language assistant with full working and inverse problem, tool catalogue, levels.
bsi-chi-esame.js	98 kB	Chemistry exam from basic to expert: 40 problem models, solved papers.
bsi-fis-elettrodi.js	14 kB	Electrode kinetics (Butler-Volmer, Tafel) and transition-state theory (Eyring).
bsi-fis-superfici.js	18 kB	Photophysics (Jablonski, Stern-Volmer), adsorption isotherms (Langmuir, Freundlich, BET), polymers.
bsi-mat-statistica.js	27 kB	Laboratory statistics: t, F, Grubbs, Dixon, uncertainty propagation, calibration line with LOD and LOQ.
bsi-nmr2d.js	21 kB	Two-dimensional NMR: COSY, edited HSQC and HMBC, built by walking the graph over the two per-atom predictions.
bsi-moldraw.js	44 kB	Interactive NMR spectrum: zoom, assignment table, peak↔atom link, SMILES/InChI history, CSV/JCAMP-DX/PNG exports.
bsi-documento.js	23 kB	Opening of any document — PDF, images, text, Word, OpenDocument, PowerPoint, Excel — with an explicit list of what could NOT be read.
bsi-quesito.js	19 kB	Recognition of spectroscopic data in a problem's text, with each datum's provenance, and step-by-step working through bsi-elucida.js.
bsi-spettrolettore.js	104 kB	Spectrum reader: JCAMP-DX with ASDF compression, prominence-based peak finding, IR band assignment, neutral losses.
bsi-mol3d.js	42 kB	3D molecule viewer: formation animation, atom selection, measurement of angles, lengths and dihedrals.
bsi-lingue.js	255 kB	Internationalisation engine: fourteen languages, writing direction included.
bsi-molingue.js	54 kB	Chemical languages: twenty-five outputs from SMILES, with IUPAC naming over a declared class.
bsi-pannelli-lingua.js	66 kB	Panels for languages and chemical languages.
sw.js	9 kB	Service Worker: offline operation and network strategy.
rdkit_lab.html	260 kB	Cheminformatics laboratory.
astro.html	2660 kB	Astrochemistry module.
chimorga.html	194 kB	Organic chemistry module.

File	Size	Role
proxy/spectra-proxy.js	10 kB	Optional proxy (Cloudflare Worker) for the AI calls.

Digests

File	SHA-256
index.html	edfc0bb143be1e408a037c80f3f46fe01d49c78f0aad79210d5677f295926165
bsi-ai-hub.js	1bc2fd5a625b8ea2f8fa488a2ed79e7fa52e2d969998a732e408ca30dd3a10c8
bsi-spettri.js	be757f15a70f884e3374d401596e2dfd18bcc852cafbb4584920e2b2d24eb3c9
bsi-cheminfo.js	59a3111b3b9436add5376fde1be6f90ded3dc08c095cebdc04da50c7f8268bbc
bsi-pretsch.js	27f727240363579fba3dba50ab3b567fbcdb26a8a9a06c0e43968b195c479db3
bsi-nmr.js	881917dbb141dfcbab9a25b9c518eaa4a8c1c52dc1020cbac90985a1cf859b72
bsi-geom3d.js	a616b91efa8f790388c99f22cc9f2d48630cb760d4f7fa87f92b99b9db106054
bsi-fisica-comune.js	8ec1678339245dcbd3f416807d0840bc8b5caebe6f99c39a510bbffa68353584
bsi-fis-schrodinger.js	b032084511102c7231bcb244b5bb647128a080a18e9e6d706ec4c727ac29d17d
bsi-fis-hartreefock.js	8210d2a1501fe6e3f57c0ce8a0e6d5d78432274792833d50bc2e821fc7566fde
bsi-fis-rotovib.js	428182687a76111f013986127ee49beb7393498387b146bd0694ff16bea7d909
bsi-fis-termostat.js	0bb9f04d7d5b144791764fa4efd9f8e658be5b5d78b40f94e18752c2366589a9
bsi-fis-dinamica.js	f1d76c737ce7f75cdc9c75665194180a4cdf401e47a105583838f31c85c9b906
bsi-fis-matematica.js	616a2d4f338fe1d371f5ba3fd4fe569e94b2d94626a8d323bba8ae58f8550f62
bsi-fis-cristalli.js	f5283d63ff075ddc19f9d200194528366f38294909db5872e7604b46e0a7d7d5
bsi-fis-nucleare.js	8ae148a2f37a40f6902faa4148e44468b4d659b6c7d1230378de92c2703a5874
bsi-mat-cas.js	eea3a1eef74d31d3fde78b7e4a2795169e8af0f357edc10050eb98d8eaf386ac
bsi-mat-basi.js	db30e16c58bde3638f13de5c3311a00598628676b7b418dc023b175efa77c214
bsi-mat-funzioni.js	68f8780d4fb4fcda6f028e4749adef333253efbf270d0caab47def5bdf30c66c
bsi-mat-calcolo.js	c3363448c4d773d29c4f680eddb79e9ad55a5eb87b033a728754df8cbee2a631
bsi-mat-lineare.js	da33ea18f2b14c21c6e630be4a1ccfe2e35e911489bdd4ededd60c0f533ac94f
bsi-mat-ode.js	698725d49dc2bf862161914e41d38c3fafd33069021eae3b11b4973c146b52ce
bsi-mat-multi.js	1040861c3b9d340febe8c4a78d212b168eb42752ee22589facf8566a401234d7
bsi-mat-complessi.js	52a2ac5594a697d3a008d9fd629bee60fcff96ea76aa16c2ccda63eafd43f316
bsi-mat-risolutore.js	709349e1d81f03506c02b9b05a060d8d3801dc37183c0e3894cc93c0aafe9f0c
bsi-mat-esame.js	de4af253244bd7d610b1a3607297d1678a529504afa23d7123aebe042e1f2c27
bsi-fis-esame.js	c9ffb883e94a07b7cda3777b34d6141bd4c47bb3c9857218a9a37f65f8728947
bsi-percorsi.js	950c686202f2934009827bc300c787063b325528b7d5ab5385a8ab3b6706345c
bsi-mat-probabilita.js	4518252496dfee0770f9e939ef2de831ac463d2d994d96d40776c59f10cd417b
bsi-mat-visiva.js	cb624f3e1e6c6df94c287f961d0f21a9d8ec852e8245ff5aeed09a083f2c5b14
bsi-fis-em.js	b20c7103f1253c76691c7cf91eae6c83491148f0aeb0928a78495a3dba270c7c
bsi-fis-suono.js	fead9e5dde961fc56c3a649cce6e8eed23046f327c8392dbfd8de5024cf0a1bf
bsi-fis-solido.js	5b6a023a84434f6fd52167e1de7a025342915d7d3287173db89901b503641cd4

File	SHA-256
bsi-chi-distillazione.js	8d13592cf7523c1dd159d3e954947a4956e44539308312dc97cd3588207fcfc0
bsi-chi-voltammetria.js	4db1a71e0ac3569b821f6cc5280470453f3a4df5c62b9695a3339367c5aa3e41
bsi-mat-numeri.js	f30d041cd9963342aa476e128e4b29d543e113d3483253377ea486de3d46484f
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bsi-mat-serie.js	722b1a1b7397b7710689af555d1fd60de73ee405f5061819d9e59d7f4631d060
bsi-mat-variazioni.js	067d676d0c04d00704b6b9feea4291629feec132d78270569327fe5afb3bb29a
bsi-mat-topologia.js	6690c261fc8b584c399c7af58f6ea2f96a1e9c56da97a0afdc281e7cc69ff14e
bsi-mat-algebra.js	59d66b5c6be431f49e5808e5c78758a7ca3cc33fba6155e56457f13dec14e237
bsi-mat-giochi.js	d1ae8048acab5a518f1ca9c9ef31e4c9346459c7a0d83ec935733a57465c4b72
bsi-mat-logica.js	26cfc591156a6c232dd40938252ce238374a6a678f08557d3bcfa899a5d0785e
bsi-mat-inferenza.js	4dadc036537e3833e396ab5dfa9968f0ccab398ccc18959ab474e6652d27e43c
bsi-mat-noneuclidea.js	4f5751bc0fbd2a1813797656f752fa16760e30032dbbdc4e1695596ec2ea8849
bsi-fis-gravita.js	8ebd280a63f5b80370cc6a9330787ae1bf3058082713f7340f17ac5bc4e4d5d5
bsi-fis-calore.js	81b9c1c60c7ba5a022f2f95e36ac17274ed1e2027d0ba36bcc24f8d6243ad3b8
bsi-fis-ising.js	6ff592edd42ca364589ec0ab1670fa9e502c445042959898e994073a6d6b7161
bsi-fis-particelle.js	0ea36a93387c4000b08ca834a6bdd82f7b232009e84b1b9c9b6ab99726dcdb63
bsi-fis-analitica.js	01b4178998372c6db7d18a312c5a5ed1c8c26267bd0b72ae63ad6fbf92ba2895
bsi-fis-laser.js	8f876f1552b50b22643526a2d16a4a8b0c0bbf5b8d9161d325f38efd16c917fa
bsi-fis-clima.js	8e136c02f7559cd22d57ccd857d06f8d6a6a89b2f38790d2782e6a45a7f2c822
bsi-chi-fasi.js	afa26d00f28caa1d462c5a8cfd74ca33e4595b5cf0b8214ece3d6ddc12140f21
bsi-chi-pes.js	528b9e19a40cdbe601f45d29fb38e71dfdb8a00ccd8360a0abfd320d1a1d01f4
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bsi-studio.js	62d001d311fa5a54df8807734bd1c2b389c730b8e6c8a6eb0f29a04d8982f410
bsi-studio-fisica.js	973bab09dad47980f0d411aef35ff0378f2266030c3fcb14e1cefe64eb63531b
bsi-studio-matematica.js	3a2d88a7f9cea52bc5c46bf86018904b2d032320726988759b3e0fb84110c5c7
bsi-risorse-materie.js	03e2fa89af1157985cdab1a08c60153ed57d8c648a5f7b67cd3f5d910c43e417
bsi-astro-sistemasolare.js	d59c52b7135d020d67a8163e9bc4d0cecea2446df93aa66389f0477a2a908ef7
bsi-astro-orbite.js	32c8123ac5a7b39a9a4a816713d0305a9d90ddb8fb714614fcbba3eb3e834c4b
bsi-astro-matcelesti.js	b2ee40c8a53fb25baf42020693539d91dc5278a14436115c69643470d57cda56
bsi-astro-stelle.js	84064a5bdf8fdc1111011201cf97a11a47396acbd3c7442d52e2c5ba74a388a3
bsi-astro-buchineri.js	c75a84cba9be45c5fec1b4f77f4ca79f9d898067006a3c309d951147ddde083c

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bsi-astro-cosmo.js	356152c5251593c68e8dd66b1182f5ee3f7c5f4f1cb26c722bdd384f8fe66d47
bsi-astro-cielo.js	26b3e33cefe7d1606d0abf8d47be6d7d9b412891ab193d386835d4e40f9fdcf6
bsi-astro-esopianeti.js	5a540d026927fc13f0081f3ba4cbfcb6de4c44056530571c77574e71f9003a49
bsi-fis-spin.js	a24dc05385fbff4826a02f95315ae1d8c57c8ee555da868a08620f24127fe3ad
bsi-fis-statistica.js	b4fd532901c0ed72d0d104c85d686d534cc90b59eacd8eaaed98255cfea553b2
bsi-fis-nuclei.js	3f393cf853d4171beb53153fa2d03da2f526ba40af2b9ad1ce4c4e426678d4db
bsi-fis-materiali.js	b2869b83fa17200da5f3900abf6959214df0dd6f88c4d8d02be88d5a6d832269
bsi-fis-elettronica.js	8620e8ba116f1d3edb3a487a7bf2aab869fbf1afc2230462e251062e7a2bf61d
bsi-mat-edp.js	525fae4f04fcc0ec7884ebba30e47d013c972a5ff77b1088ab4bca41bd04a50b
bsi-mat-analisicomplessa.js	0eb6e5181471af7e85eb3ee42012daddbe1fdac932c0c4c28cf67135875302d8
bsi-mat-markov.js	0e122fac691f1faa2e464214a7d24c9a687b16491c5964b760e0df191b13a35b
bsi-mat-svd.js	ce6d77e921c469c532f9f6de2689f7b2a8c8de656f18ee32b1fc4e1977538931
bsi-mat-codici.js	1be8fa29332df98115db6402881d621c1806599129aa279ee5678ef586445795
bsi-spectra-live.js	1ea2fe073f0b0e8ea5690c2802e26198c83ac8539eaa609754dd18895e56a464
bsi-spectra-androide.js	90ee017923c2899233cd1950e9d63c22dcd8d9daf0fcfc711e3482aa5b6dfcdf
bsi-spectra-mente.js	bd2bd20245ee552520fdc3af471a5850aec2a72c40e0d6d807211c2464e0cf34
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bsi-aggiorna.js	5950d90103a7d603175e4b577e2394bb9a6d006a4ff22e48d4a1cc70661f1b8e
bsi-fis-fluidi.js	65d94ea45133f9537fd3af0feb96b12539a866cde8c12a29cb48c5156b001b9e
bsi-mat-vettoriale.js	fc3116f45b849118cf41c23f13cc2c07f1d9bdd03341baf6cc81b152a5b6225e
bsi-chi-cinetica.js	8ba3ad927e4a1858779ffd8ba798c1211a844475b1ad4b571cd1acfd41ea30b0
bsi-fis-meccanica.js	07d67eb501958079347fa1905a9c212ca2a588c5d1930f615bd0bd07b36a9279
bsi-fis-onde.js	23bc4412088de214ad1f6f2671c2507eadd08ff252f21966f557c31ed6dae073
bsi-fis-circuiti.js	098679cdbb52241b6035e23076c82a25fe3d9a72b6f5bf7d4f880f887223f375
bsi-fis-termo.js	c04fc820b4d874b41b18407b200410d8b8862b3e20f8813660ad009f6af67d3b
bsi-mat-numerici.js	8656bdf657dcc3468f78951ed3d8e1056d953b147e04595c6bfaad647fa8ca93
bsi-mat-geometria.js	609bf6c951d438aa0b0639fd2e736a52b340a363294ee564acfe9510966334be
bsi-chi-equilibrio.js	de2368be8299a4ebdce9401b77c7bea390208f1372460f17f033e27d0ead88af
bsi-lab3d.js	9b1424f3ad744cc2abd42be04f9db0907578452ecb66156e2c809a5c774a1604
bsi-vista-molecole.js	4a7eb7cd7d8a0033ad0381d936d8e36b2b2e6b75af04988ecc5e9f4e8022cbc4
bsi-assistente.js	00053872ed83218c81b8867b075a5142c6108dbc1f41e7693df1e9aa0897c4f8
bsi-chi-esame.js	ddb5d73fcf887e03770b3f0794fecaa2f8c918f4fae445bf57ff7a637f4643e7
bsi-fis-elettrodi.js	249bb92ac39b78e916eaa0ecf80a2b1aac5d10249bb6b51628f61839c4b64240
bsi-fis-superfici.js	3daec64d2c4755e70279b8ef2f433e4c98d6e90ff7d8c7d9c0689a24f080abd8

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`bsi-mat-statistica.js`	`0b0c82afa13173b84308589cf16fbaa8b30c5e575fd8fdee00034b70c34d17fc`
`bsi-nmr2d.js`	`a0b1db87f27ffa8c124d617e4903f61e35b89c30c35162527ad023325c03fa78`
`bsi-moldraw.js`	`45cacdca20c8091cbfb32e256c179de8c32c71068a5d1008fbef9be005b5df0e`
`bsi-documento.js`	`0e9ad6043a396652549b30879764cc401f8b7bd25689b5bb5f37cf26f7d7d8d9`
`bsi-quesito.js`	`6f686abcfdc3959d8c865a3e6ee7129224de52992ab1ba3da07372a986769cdf`
`bsi-spettrolettore.js`	`14daf41305b0a7271c6ff5f7c9078a4e5f0e0149f32460a9608ed1996f810256`
`bsi-mol3d.js`	`93fea4a6811ba4263de450db4e7ae1801d4fde196cf7683d0100846c51751893`
`bsi-lingue.js`	`f08f2a9bd754d9212d69a2a1f8169c0a24ee992ec2af1bcaf54c4ca37b10d229`
`bsi-molingue.js`	`e45fec1ee20e3e7b9a7bea3d6e20aa4e33e74ec43578bfa7bfb94646b7bc53fb`
`bsi-pannelli-lingua.js`	`4af94d70ef104cf21be18ff7fbc7a2248c56e376e2ae48e8cac83878f4554f93`
`sw.js`	`6dc41fca07ff8e6d16ade0ec93989f6e3890378717702f87a677da09fea1e998`
`rdkit_lab.html`	`066a42092b829df2fa306b691b9d162b6d44edfdb5317ce4cc8a276bc34c80ad`
`astro.html`	`3376b2b7f26ed483f2d698c6692b31f21c9f245f87d1fb1884216fb8b1da489d`
`chimorga.html`	`047934abc73a79fbaf06ed7036b6e89a913ebdf5354fc52cecdbee7e8f94f51d`
`proxy/spectra-proxy.js`	`808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962`

3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

Package	Version	Licence
`playwright-core`	1.62.1	Apache-2.0

4. Runtime dependencies

None. The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

Service	Purpose	If unreachable
PubChem PUG-REST	IUPAC name, CAS, GHS pictograms, 3D conformers	the app uses local data and says so
NASA / ESA	astronomical images and data	the module shows the tabulated data
AI providers (user's choice)	assistant answers	the assistant reports it and offers alternatives

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage`, or on the user's own proxy (see `proxy/`).

5. Licence compatibility

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The remaining obligations are of **attribution**: the licence notices of the upstream projects are reproduced in `THIRD_PARTY_NOTICES.md`.

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